How Scoring Responses Change Attribution Evaluation in a Diffusion Policy

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Abstract

Feature attributions rank inputs associated with a model's prediction. Their evaluation depends on how a scalar response scores predictions after features are removed or restored. For robot policies, we separate the attribution target from the scoring response. We compare squared distance (Q) and stabilized distance (N) from a fixed reference action. An identity shows that N cannot increase insertion or deletion area under the curve (AUC) when no intervention overshoots, that is, moves the prediction farther from the reference, in squared distance, than replacing every feature with its baseline. Strict decreases make an unchanged ranking look worse by insertion and better by deletion; absolute thresholds can give opposite verdicts. We rescore saved curves for PickCube, a simulated pick-and-place task, from a model recorded as a pretrained Robotics Diffusion Transformer. Q-ranked and N-ranked cohorts ranked features using Q and N, respectively; each contains 750 policy calls from 30 episodes with no successes. Across eight combinations of cohort, modality and direction, median Q-minus-N differences are 0.104 to 0.159, with 75.6% to 94.3% positive. Overshoots occur in 16.0% to 74.0% of calls. Extreme calls, mainly each episode's first prediction, make vision-insertion means strongly negative. An example shows that target choice can also reverse rankings. Separate numerical diagnostics show that integration error can change evaluation results or leave them unchanged. A smooth counterexample shows that agreeing uniform integration grids can give incorrect coordinates. We recommend specifying targets and responses separately, comparing rankings under matched controls, reporting full distributions, and checking numerical sensitivity.

Keywords: integrated gradients, attribution evaluation, diffusion policies, perturbation metrics, numerical integration, robotics

1 Introduction

Attributions assign image regions, language representations or state coordinates contributions to a robot policy's prediction. A persuasive ranking need not identify inputs that meaningfully affect actions. Diffusion policies predict action sequences by repeatedly refining noise conditioned on observations[1]. Their vector outputs require a scalar attribution target, such as discrepancy from a fixed reference action[2].

Q measures squared distance and N stabilized distance from the reference; "Q-ranked" means ranked using target Q. Deletion area under the curve (AUC) summarizes responses as features are removed, with lower conventionally better. Consider removing the highest-ranked image features. In Q-ranked curves, median vision-deletion AUC changes from 0.448 scored with Q to 0.291 scored with N. An illustrative, unvalidated absolute bar of 0.40 rejects the first and accepts the second. Absolute thresholds, cross-response comparisons and assumed 0.5 references therefore need

response-matched calibration. Only scoring changed; rankings, interventions and predicted actions stayed fixed.

Integrated Gradients (IG) adds model sensitivity along a straight baseline-to-input path[3]. Completeness means coordinate attributions sum to the target's endpoint change. Three questions differ: what the scoring response measures, whether integration recovers the intended attribution, and whether the ranking is useful. We study the first two using the Robotics Diffusion Transformer (RDT)[4].

Insertion and deletion scores evaluate a ranking by progressively restoring or removing features[5]. Their interpretation depends on the interventions: retraining after removal changes the question[6], masks can reveal information[7], and perturbation metrics can disagree[8]. Hama, Mase and Owen established that response scale and feature interactions affect insertion and deletion evaluation and its random-order reference[9]. Here we separate target choice from scoring response for vector-output action discrepancies, derive their response-only ordering, and measure the effect and normalized-AUC instability on saved robot-policy curves.

The integration budget is the number of intervals used to approximate IG. Earlier work finds instance-dependent completeness errors as this budget changes[10]; IDGI reduces explanation noise by retaining the component of each integration step along the gradient[11]. Our concern is recovery of the specified IG integral. We use quadrature aliasing, where sampled nodes miss variation between them, to examine what finite numerical checks can establish.

First, scoring with N instead of Q cannot increase either AUC for in-range curves, whose saved residuals all lie between zero and the all-baseline value (Section 2.4). Strict decreases make insertion look worse and deletion better; saved curves measure their size and prevalence. Second, overshoots beyond this range can dominate means through extreme first calls. Third, an example shows target-dependent IG ranking reversal. Finally, real-model diagnostics expose numerical problems, while a counterexample limits agreement and completeness checks. Section 4.1 gives reporting steps.

2 Methods

2.1 Research design and policy setting

We combine algebra, saved-curve rescoring and separate numerical diagnostics. A policy call predicts an action chunk; the first call is index 0 of each episode. Target and scoring response are separate choices (Figure 1); rescoring requires no new policy execution.

Attribution covers 64 predicted timesteps and eight action dimensions, so $d_a = 512$. RDT places each robot's actions and states in a shared, padded 128-dimensional representation; "active" coordinates are the eight this robot uses, and the active action chunk consists of these eight coordinates at every predicted timestep. Our ManiSkill3[12] evaluation executes every fourth action: 16 per call and 25 calls give 400 environment steps. The headline task, PickCube, moves a cube to a target. A separate numerical collection covers PickCube (cube placement), StackCube (stacking), PegInsertionSide (peg insertion) and PickSingleYCB (household-object pickup).

2.2 The function being explained

Fix context c: model, diffusion noise and other modalities. Let $\mathcal{M}$ denote vision, language or state, with input representation x and baseline x'. Let $g_{c,\mathcal{M}}(u)$ predict the active action chunk after replacement by u, with reference $a_c = g_{c,\mathcal{M}}(x)$ computed once.

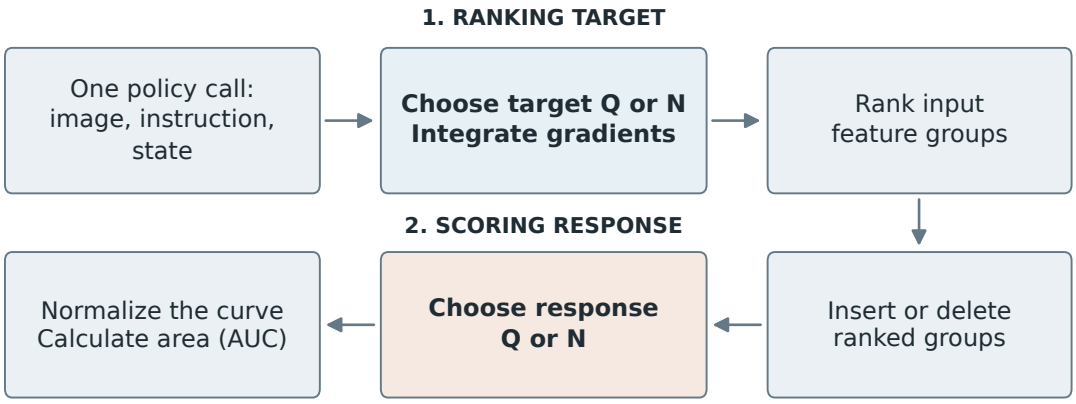


Figure 1 | Ranking and scoring are separate choices. Q is the quadratic action-discrepancy function and N its stabilized norm counterpart (Equation 1). IG ranks feature groups using one target. The scoring stage covers both the interventions and their response curves: either response can score the same predicted actions. Rescoring keeps the observation, reference action, diffusion noise, ranking and interventions fixed.

Define

$$\begin{aligned} r(u) &= \|a_c - g_{c,\mathcal{M}}(u)\|_2^2, \\ Q(u) &= -\frac{r(u)}{2d_a}, \qquad N(u) = -\sqrt{r(u)+\epsilon}, \qquad \epsilon = 10^{-12}. \end{aligned} \tag{1}$$

Either function can be the attribution target or scoring response. Both reward proximity to the reference in action coordinates. Q is an auxiliary discrepancy score, not a policy log likelihood. Q's magnitude grows with squared residual, whereas $|N|$ grows approximately with its square root when r is large compared with ϵ. At the actual input,

$$Q(x) = 0, \qquad N(x) = -\sqrt{\epsilon}.$$

Both gradients vanish there. A positive constant multiplier changes neither exact absolute-coordinate rankings nor endpoint-normalized AUC.

Fixed noise makes the attributed mapping one function along the path. Redrawing noise would change that function. Averaging over noise requires a different target and sampling procedure; interpreting the discrepancy as a distance between action distributions requires further assumptions[2].

2.3 Features, baselines, and Integrated Gradients

RDT encodes inputs as tokens; a group contains coordinates replaced together. SigLIP image and T5-XXL language encoders feed adaptors mapping outputs into policy representations[4]. Vision attribution follows the adaptor, with 729 patch-token groups from one external camera. Deleted tokens receive adapted tokens at matching positions from a uniform gray image. Five other camera slots remain fixed background images.

Language groups are token positions retained by the attention mask after the encoder and language adaptor. Their count was not preserved in the saved curve records. A deleted position receives its matching adapted baseline token. The baseline encodes padding and end-of-sequence tokens, pads the encoder output with zeros, then applies the adaptor, so its adapted value need not be zero. The encoder is not rerun after replacement; other tokens can retain information from the removed

position. Baselines specify a comparison rather than a uniquely neutral input[13], and straight paths between language representations need not correspond to valid text[14].

State uses the eight active coordinates of the normalized 128-dimensional representation, each its own feature group, with a zero baseline and differentiation through its adaptor. Saved PickCube rescoring covers only vision and language; separate numerical diagnostics include state.

For target $F \in \{Q, N\}$, coordinate i has attribution

$$A_i^F = (x_i - x_i') \int_0^1 \partial_i F(x' + \alpha(x - x'))\, d\alpha. \tag{2}$$

Under the chain rule and path integrability, completeness gives $\sum_i A_i^F = F(x) - F(x')$. This total is $r(x')/(2d_a)$ for Q and $\sqrt{r(x') + \epsilon} - \sqrt{\epsilon}$ for N. Vision and language groups are ranked by the sum of absolute coordinate attributions within each group.

With m intervals, $m + 1$ evaluations and $f_i(\alpha) = \partial_i F(x' + \alpha(x - x'))$, the saved PickCube runs used the endpoint-inclusive average

$$A_{i,m}^{\text{avg}} = (x_i - x_i') \frac{1}{m+1} \sum_{j=0}^{m} f_i(j/m). \tag{3}$$

The numerical sample uses the composite trapezoidal rule,

$$A_{i,m}^{\text{trap}} = \frac{x_i - x_i'}{m} \left[\frac{f_i(0) + f_i(1)}{2} + \sum_{j=1}^{m-1} f_i(j/m) \right]. \tag{4}$$

The trapezoidal rule is second-order accurate for sufficiently smooth integrands[15]. The endpoint-inclusive average equals $m/(m+1)$ times that estimate plus $(x_i - x_i')[f_i(0) + f_i(1)]/[2(m+1)]$, giving a generally first-order endpoint correction. Here "order" describes how error decreases as the interval count grows. Positive ϵ makes N smooth for a smooth action map, but its narrow transition near zero can remain unresolved at practical budgets. The saved runs used $m = 64$, brain floating point 16 (bf16), and zero replacement of nonfinite gradients; the replacement frequency was not recorded. The implementation for the saved PickCube runs and the numerical collection specifies five-step DPM-Solver++ sampling, with gradients through every denoising step and noise held fixed within each attribution. The saved PickCube curves lack run configurations; the Q-ranked episode records store five solver steps, and for the N-ranked runs the step count is inferred from the default configuration.

2.4 Interventions and scoring

We compare responses on the same interventions. Let G count eligible groups (the camera patches, attended language positions or, in the numerical sample, state coordinates defined above), and let P_k mask the first k ranked groups, expanded over their coordinates. With $\odot$ denoting coordinatewise multiplication, insertion and deletion inputs are

$$x_k^{\text{ins}} = x' + P_k \odot (x - x'), \qquad x_k^{\text{del}} = x - P_k \odot (x - x'). \tag{5}$$

For response $S \in \{Q, N\}$, endpoint normalization gives

$$C_S(u) = \frac{S(u) - S(x')}{S(x) - S(x')}. \tag{6}$$

This assigns zero to the all-baseline input and one to the actual input. Self-reference gives $r(x) = 0$ by definition; we check it because records were stored in finite precision. Write $B = r(x') > 0$ for

the all-baseline squared residual. An intervention is in range if $0 \le r(u) \le B$ and an overshoot if $r(u) > B$. A curve is fully in range if every saved point is in range. An undershoot below zero is impossible in exact arithmetic; its check guards against roundoff.

Area under the curve (AUC) integrates normalized values over the fraction of groups intervened on. Higher insertion AUC and lower deletion AUC are conventionally better. The saved nominal fractions are 0, 1, 5, 10, 20, 30, 50, 75 and 100%. For $0 = q_0 \le \cdots \le q_J = 1$, with $J = 8$ fraction intervals and integer group count k_j at fraction q_j, we use

$$\mathrm{AUC}_S = \sum_{j=1}^{J} \frac{q_j - q_{j-1}}{2} \left[C_S(x_{k_j}) + C_S(x_{k_{j-1}}) \right]. \tag{7}$$

Actual fractions k_j/G can differ from the nominal schedule; the realized masks and rounding rule were not preserved with these curves. In all 3,000 language curves, the saved response values at 1% and 5% are equal, without establishing mask identity. Both responses use the same nominal AUC weights. Dropping the 1% node moves median language paired differences by at most 0.004; for fully in-range calls, the paired change is bounded by 0.005 (Appendix A.1).

The paired response effect is

$$\Delta = \mathrm{AUC}_Q - \mathrm{AUC}_N. \tag{8}$$

Positive Δ means scoring with N lowers that AUC. Ranking, interventions, predicted actions, baseline and schedule remain fixed within each pair. The median of paired differences need not equal the difference of the two marginal medians.

2.5 Data sources and claim dependencies

The saved PickCube runs contain response curves and episode records. One cohort used Q to rank features and the other used N; we call them Q-ranked and N-ranked. Both are recorded as the same pretrained RDT-170M without PickCube fine-tuning; the original checkpoint file was not preserved. RDT-170M is the publicly released 170-million-parameter small variant of RDT[16]. RDT-1B names the larger, billion-parameter model[4]. Each cohort contains 30 episodes with 25 calls, using environment-seed blocks 42 and 142 with 15 episodes each. These are environment seeds, not training runs. Every episode reaches the 400-step, 25-call cap without success.

Shared environment seeds and episode indices give the same seeded initial-state setup. Yet vision B differs in all 750 aligned calls. Its cross-cohort log correlation is 0.992 at call 0, drops to 0.289 by call 2, then varies between about 0.4 and 0.8, ending at 0.66 at call 24. Similar vision summaries thus come from related trajectories. Missing original contexts prevent assigning differences to particular inputs or noise. The cohorts support separate response comparisons under different ranking targets, not a comparison of ranking quality.

Both full-episode PickCube cohorts passed the rescoring procedure described below. A separate short verification set, distinct from the numerical collection, retains only calls 0–3 (0–6 for PegInsertionSide) on four tasks and is not rescored because it covers only early calls, a different population from full episodes. Full-episode $m = 128$ PegInsertionSide/PickSingleYCB and fine-tuned RDT-1B runs each use one ranking target and remain unprocessed by this procedure. A single target still permits rescoring, a direct next test. RDT-1B language baselines differ across seeds, so only their vision curves are comparable across seeds. Records from a third ranking target, maximum coordinate deviation, are excluded because their stored scores cannot recover squared residuals. Table 1 links sources to claims.

Table 1 | Data sources support distinct claims. Saved PickCube runs supply both ranking cohorts. Success refers to the robot task. Numerical diagnostics test the separately specified offline function.

Source	Inputs and scope	Supported claim and dependency
Analytic results	Defined functions, paths and integration nodes.	Response ordering, target reversal and limits of sampled integration under stated assumptions.
Saved PickCube runs	Each cohort: 750 calls, 30 episodes, two environment-seed blocks, zero successes. RDT-170M, $m = 64$, bf16.	Within-curve response effects and overshoots from saved response values, endpoints and nominal fractions. Original noise was not preserved.
Numerical sample	Separate collection capped at 13 calls per episode; 12 completed contexts from 6 episodes across four tasks with RDT-170M.	Repeatability and budget sensitivity of an offline fp32 function, using this collection's preserved noise and encoder outputs.

Each cohort stored curves scored with its own ranking function. We recover $r = -2d_aQ$ or $r = N^2 - \epsilon$ from those values, then recompute both responses, endpoints and AUCs. N-ranked actual-input endpoints store approximately -10^{-6}, whose inversion gives a tiny negative residual from roundoff. Only this actual-input residual, zero by definition, is set to zero; Q-ranked endpoints give zero directly. No intervention point undershoots, and all calls have $B > 0$. We remove 24 duplicate records that differ only in timing, leaving every distinct saved call.

2.6 Descriptive statistical analysis

The response identity predicts a nonnegative per-call difference for fully in-range curves. Medians and positive-call shares describe the typical effect, while means retain its tail magnitude. We chose this emphasis after examining the data. We report all eight combinations of cohort, modality and direction, which reuse observations and are dependent.

Descriptive 95% percentile intervals use 10,000 whole-episode bootstrap samples with seed 0. Equal-episode and equal-call means coincide because every episode has 25 calls. Intervals describe logged episodes, not independently trained models. We split means into AUC-weighted in-range and overshoot contributions and count single-episode omissions preserving their sign. No finite AUC or paired-difference values are clipped or trimmed.

2.7 Numerical diagnostic procedure

The numerical sample tests a separate offline RDT function recomputed in 32-bit floating point (fp32) from cached encoder outputs. Weights and adapted image/language values retain bf16-rounded values lifted to fp32; state is readapted in fp32. Its source is a separate collection capped at 13 policy calls per episode, which preserved noise. A new reference action is computed with that noise held fixed. Calls 0 and 12 were selected prospectively to cover the initial and later portions of this short collection. Of 24 planned contexts, including 8 for RDT-1B, 12 are complete, 9 unfinished and 3 have selected calls that were never reached.

IG budgets are 128, 256 and 512 intervals for vision and language, and 256, 512 and 1,024 for state. Finest maps were computed twice; gradients were evaluated three times at seven fixed path positions. Equality checks compare repeated gradients, maps and responses; coverage checks require the planned positions, budgets, repeats and response comparisons to be present. These are dependent checks on the same contexts. Using the trapezoidal estimator $A_{i,m} = A_{i,m}^{\text{trap}}$, the relative completeness residual E_m and relative coordinate change D_m are

$$E_m = \frac{|\sum_i A_{i,m} - (F(x) - F(x'))|}{|F(x) - F(x')|}, \qquad D_m = \frac{\|A_m - A_{2m}\|_1}{\|A_{2m}\|_1}. \tag{9}$$

Comparisons cover all three pairs of budgets: the two adjacent pairs, as in Equation 9, and the smallest with the largest budget, for which the largest budget's map replaces A_{2m} in D_m. Zero denominators are undefined. The numerical sample also scores curves with the root-mean-square action discrepancy $\text{RMS}(u) = (r(u)/d_a)^{1/2}$, whose endpoint-normalized curve, when $B > 0$, equals that of N with $\epsilon = 0$. Absolute endpoint changes at or below 10^{-12} for Q, or 10^{-8} for N and RMS, are flagged near-zero. A budget selected for final use would require a relative completeness residual at most 1% when its endpoint change exceeds the cutoff; none is selected here. Mean path-gradient maps omit the IG factor $x - x'$. Both map comparisons require relative coordinate L_1 change at most 1%, Spearman at least 0.99 for nonconstant group scores, and top-five-percent overlap at least 0.95, replaced by matching top-one for an empty subset. Maximum RMS-curve and absolute raw RMS-area changes must be at most 1% and 0.5% of all-baseline RMS, respectively. Absolute normalized-AUC changes must be at most 0.01 under each common Q, N or RMS response when its endpoint gap exceeds the cutoff; Section 3.5's examples use Q. These are sensitivity criteria, not exact-IG error bounds. Budget-check violations count all failed comparisons and recorded completeness-threshold failures, including at coarse budgets. Finest-pair counts include a context when any IG or mean path-gradient comparison between its two largest budgets fails.

2.8 Ethical considerations

This study used simulated robot policies and saved numerical records. There were no human or animal subjects.

3 Results

3.1 Scoring with N instead of Q cannot increase in-range AUC

For squared residual r, Equation 6 gives

$$\begin{aligned} C_Q(r) &= 1 - r/B, \\ C_N(r) &= \frac{\sqrt{B+\epsilon} - \sqrt{r+\epsilon}}{\sqrt{B+\epsilon} - \sqrt{\epsilon}}, \\ C_Q(r) - C_N(r) &= \frac{(\sqrt{r+\epsilon} - \sqrt{\epsilon})(\sqrt{B+\epsilon} - \sqrt{r+\epsilon})}{B}. \end{aligned} \tag{10}$$

The difference is positive for $0 < r < B$, zero at endpoints and negative at overshoots (Figure 2). Thus N cannot increase either fully in-range AUC, and the decrease is strict if some point with positive AUC weight has $0 < r < B$. Curve monotonicity is unnecessary. As $\epsilon \to 0$, setting $t = r/B$ gives $C_Q = 1 - t$ and $C_N = 1 - \sqrt{t}$. More generally, an increasing convex rescaling lies below the normalized endpoint chord in range; N is such a rescaling of Q. A strict decrease makes the fixed ranking look worse by insertion and better by deletion.

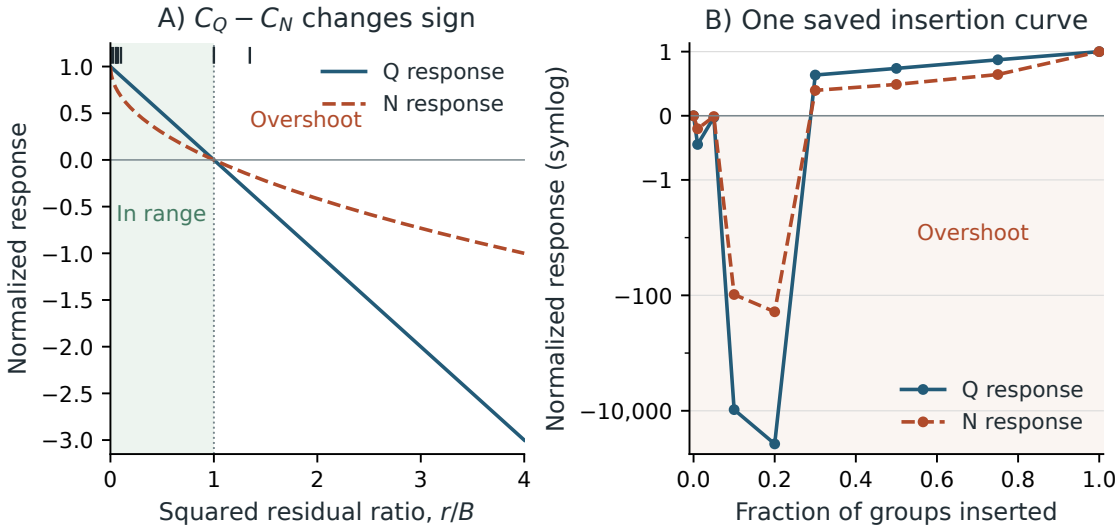


Figure 2 | The response difference $C_Q - C_N$ changes sign at overshoot. A) Normalized responses versus r/B, with $B = 1$, $\epsilon = 10^{-12}$. Rug: saved nodes from the Q-ranked vision-insertion call nearest the median difference. **B)** Extreme call from Section 3.3; negative values mark overshoot. Blue: Q; orange: N. The symmetric logarithmic scale is linear for $|C_S| \leq 1$.

The pointwise difference is at most $1/4$, including at overshoots, which only lower it. Consequently every defined saved curve satisfies the pointwise-derived bound $\Delta \leq 1/4$. On the nine-point schedule, zero endpoint differences carry 13% of the AUC weight, giving the tighter schedule bound 0.2175 (Appendix A.1). Fully in-range curves additionally have $\Delta \geq 0$. These results hold for any fixed ranking, independently of how accurately IG was computed.

A random-order reference is the expected AUC of a random feature ordering, often assumed to be 0.5. Consider an additive scalar policy summing n equal input features, with all-one input and zero baseline. For $n = 100$, every ordering gives insertion and deletion AUC about 0.660 under Q on the nine-point schedule, versus 0.5 under N with $\epsilon = 0$. On all $n + 1$ whole-feature fractions, both quadratic AUCs are $2/3 - 1/(6n^2)$. Squaring the discrepancy creates interactions even for an additive policy[9]. These are analytic calibration values, not measured random rankings on RDT (Appendix A.3). Subtracting them from the opening deletion medians would put both about 0.21 below their respective references. That arithmetic illustrates why an absolute-bar flip need not survive response-matched calibration, and why the actual reference must be measured for each response.

3.2 Saved curves show a substantial response-only shift

Within the Q-ranked cohort, changing only the response from Q to N lowers median vision-deletion AUC from 0.448 to 0.291. Within the N-ranked cohort, changing from N to Q raises it from 0.283 to 0.440. Both show a substantial change while the ranked interventions remain fixed.

All eight paired medians are positive, from 0.104 to 0.159, and 75.6% to 94.3% of calls have positive differences (Figure 3). In each cohort, these shifts are comparable to the roughly 0.16 shift that a response change gives every ordering of the additive calibration. Fully in-range calls account for 26.0% to 84.0% across cases; all are positive in these data. Among calls with an over-

shoot, 55.9% to 75.8% still have positive differences. These measured proportions, together with the identity, explain the positive medians.

First-call vision-insertion medians are -0.414 for Q-ranked and -0.845 for N-ranked calls, whereas both later-call medians are positive. The largest positive difference, about 0.214, reaches 98% of the schedule bound. Among fully in-range Q-ranked vision-insertion calls, the middle half of paired differences spans 0.098 to 0.135: the response shift is not a constant offset.

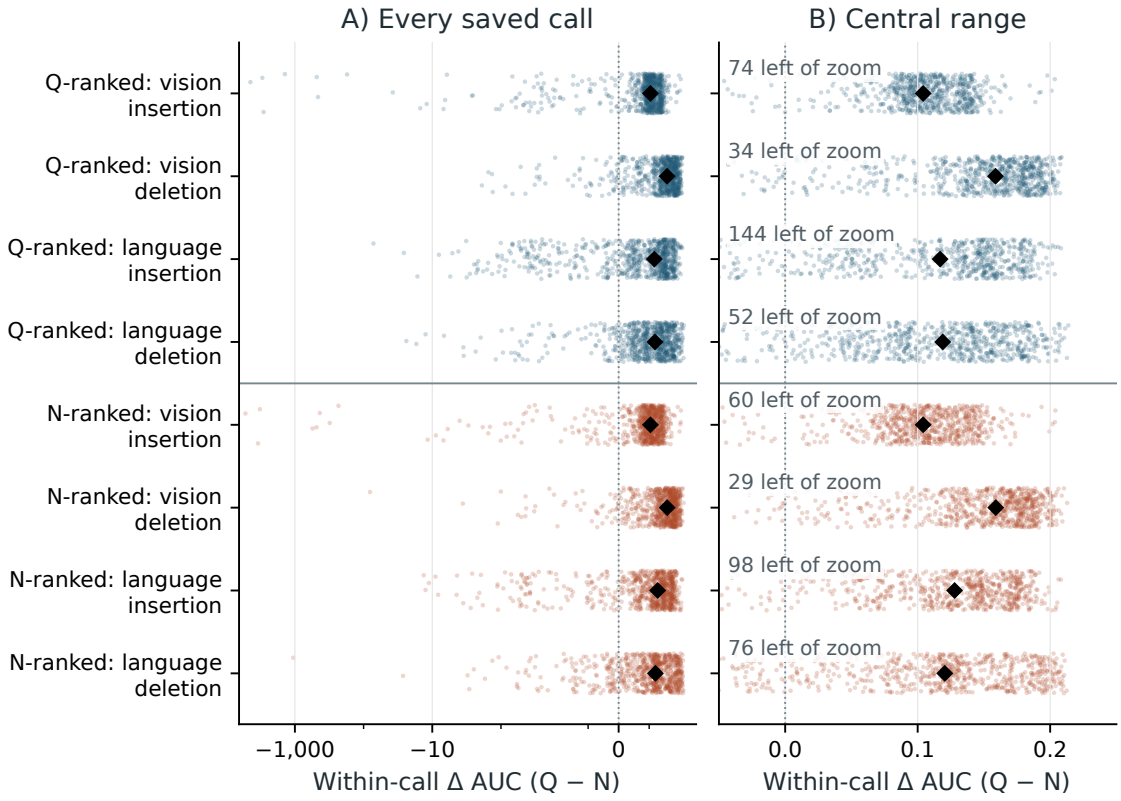


Figure 3 | Positive typical shifts coexist with long negative tails. Each row uses 750 saved PickCube calls from 30 episodes. Row prefixes identify Q-ranked and N-ranked cohorts. Positive differences lower AUC under N, making insertion look worse and deletion better. **A)** All calls on a symmetric logarithmic scale, linear for $|\Delta| \leq 0.25$. **B)** The window $[-0.05, 0.25]$; every outside value lies to the left, with counts shown there. Black diamonds mark medians; blue and orange identify Q-ranked and N-ranked cohorts.

3.3 Overshoots can dominate means through extreme calls

Overshoot points occur in 16.0% to 74.0% of calls. The in-range contribution is nonnegative by construction, but negative overshoot contributions reverse six of eight means despite positive medians (Table 2). Three negative means have 95% episode-bootstrap mean intervals entirely below zero: Q-ranked vision insertion $[-33.82, -3.97]$, N-ranked vision insertion $[-37.87, -3.17]$, and Q-ranked language insertion $[-0.405, -0.028]$. The other three negative means have intervals spanning zero: Q-ranked language deletion $[-0.102, 0.063]$, N-ranked language insertion $[-0.096, 0.027]$, and N-ranked language deletion $[-4.208, 0.078]$. Vision-insertion differences

below -100 occur in the same 7 of 30 episodes in both cohorts. These extreme episodes drive the vision-insertion mean intervals, so the two cohorts are not independent evidence for this tail.

Table 2 | Typical response effects and tail contributions differ. All cases use the saved PickCube runs and $\Delta = \mathrm{AUC}_Q - \mathrm{AUC}_N$; positive values mean scoring with N lowers that AUC. **A)** Paired median, descriptive 95% episode-bootstrap interval, positive-call share and share of calls with any overshoot. **B)** Equal-episode mean, in-range and overshoot contributions, and single-episode omissions preserving the mean's sign. Values are rounded independently.

A) Typical change and prevalence

Cohort	Input / curve	Median	95% interval	Positive share	Overshoot share
Q-ranked	Vision ins.	0.104	[0.096, 0.112]	88.1%	41.6%
Q-ranked	Vision del.	0.159	[0.148, 0.169]	92.5%	16.9%
Q-ranked	Language ins.	0.117	[0.067, 0.137]	75.6%	74.0%
Q-ranked	Language del.	0.119	[0.091, 0.138]	88.5%	37.5%
N-ranked	Vision ins.	0.104	[0.095, 0.113]	90.0%	38.9%
N-ranked	Vision del.	0.159	[0.146, 0.169]	94.3%	16.0%
N-ranked	Language ins.	0.128	[0.111, 0.145]	84.1%	65.5%
N-ranked	Language del.	0.120	[0.099, 0.131]	83.2%	41.6%

B) Means and tail contribution

Cohort	Input / curve	Mean	In-range contribution	Overshoot contribution	Same sign
Q-ranked	Vision ins.	−17.097	0.114	−17.211	30/30
Q-ranked	Vision del.	0.123	0.144	−0.021	30/30
Q-ranked	Language ins.	−0.182	0.131	−0.313	30/30
Q-ranked	Language del.	−0.009	0.120	−0.130	27/30
N-ranked	Vision ins.	−18.562	0.114	−18.676	30/30
N-ranked	Vision del.	0.018	0.144	−0.125	30/30
N-ranked	Language ins.	−0.033	0.142	−0.175	30/30
N-ranked	Language del.	−1.371	0.125	−1.496	29/30

For Q-ranked vision insertion, the mean is -17.10 but the median is 0.104. Its in-range contribution is 0.114 and overshoot contribution -17.21. Figure 2B shows an extreme call with $B \approx 0.0174$ and maximum squared residual about 653, a ratio of about 37,500. Q and N AUCs are about $-4{,}465$ and -26, producing a paired change near $-4{,}439$. The N-ranked vision-insertion minimum is about $-5{,}243$.

First calls are only 30 of 750 calls per cohort, yet contribute -16.92 to the -17.10 Q-ranked vision-insertion mean and -17.75 to the -18.56 N-ranked mean. The 720 later calls contribute -0.18 and -0.81, respectively, on the same 750-call denominator. First calls supply 98% and 95% of the negative overshoot contributions. All first-call vision-deletion curves overshoot, versus

12.5% to 13.5% of later calls. First calls also contribute -1.45 to the N-ranked language-deletion mean, while later calls contribute $+0.08$. The negative tail is concentrated in initial calls for these cases.

For scale, vision B has medians 11.8 and 12.4 in the Q-ranked and N-ranked cohorts, with 5th–95th percentile ranges about 2.7–69 and 2.9–82. In each cohort, 14 of the 30 first calls have $B < 0.5$, accounting for 14 of 17 such calls overall. The six most negative Q-ranked vision-insertion calls are all first calls. In these data, small B is necessary but not sufficient for an extreme call: every vision-insertion call with Δ below -100 has $B < 0.5$, yet only 7 (Q-ranked) and 8 (N-ranked) of the 17 small-B calls per cohort fall that low. Their largest intermediate squared residuals lie between 191 and 825, whereas those of the other small-B calls are at most 8.4. Partially inserted inputs mix baseline and actual representations and may lie outside typical policy inputs[6,7], which could produce such residuals. Why roughly half of initial states have such a small B remains open; poor competence is one hypothesis. Neither explanation is tested here.

Table 2B's *Same sign* count measures sensitivity to dropping one episode, not uncertainty or the first-call pattern. Every insertion mean stays negative under each omission. N-ranked language insertion does so 30 times even though its mean interval spans zero. Language-deletion means change sign in three Q-ranked and one N-ranked omission. Medians, signs and means therefore answer different descriptive questions.

3.4 Changing the target can reverse exact IG rankings

Changing the target poses a different problem even for exact integration. Where gradients exist,

$$\nabla Q(u) = \frac{\sqrt{r(u) + \epsilon}}{d_a} \nabla N(u). \tag{11}$$

The factor varies along the path and reweights coordinate contributions. For $g(x_1, x_2) = x_1^2 + \frac{4}{5}x_2$, baseline $(0, 0)$, input $(1, 1)$, reference $9/5$ and $d_a = 1$, exact Q attributions are $(23/30, 64/75)$, ranking feature 2 first. N ranks feature 1 first by about 0.2 (Figure 4A; Appendix A.2). In this example, Q's weight on ∇g decreases to zero along the path, suppressing feature 1's growing gradient. The factor in Equation 11 itself remains positive at the input for positive ϵ. This is an existence example; its path-weighting mechanism is specific to IG, whereas squared-discrepancy masking scores also occur in policy analysis[2].

3.5 Numerical changes can affect evaluation or escape its checks

The sample tests the separate offline fp32 function, not saved $m = 64$, bf16, endpoint-inclusive rankings. The 11,592 recorded diagnostic checks include 6,048 budget checks: 302 violated, 5,602 satisfied and 144 not applicable because top-five-percent selection yields no state groups. The 5,544 other checks all pass, giving 11,146 satisfied overall; they include 2,880 repeat-related equality and coverage checks, 1,800 of them equality checks. Among 12 contexts per modality and target, finest-pair violations affect 4 vision N and 1 vision Q contexts, 2 state contexts under each target, and no language contexts. Passing equality checks coexist with budget sensitivity (Figure 5).

Between the two finest budgets, state IG changes reach 32.1% for Q and 20.8% for N; finest completeness residuals reach 11.4% and 6.7%. Both state contexts from the only completed PickSingleYCB episode (episode 1) exceed the 1% completeness threshold under both targets, with endpoint target changes above the near-zero cutoffs; their $m = 256$ Q residuals exceed 100%. At that episode's call 0, vision changes by 2.05% under Q and 1.23% under N between 256 and

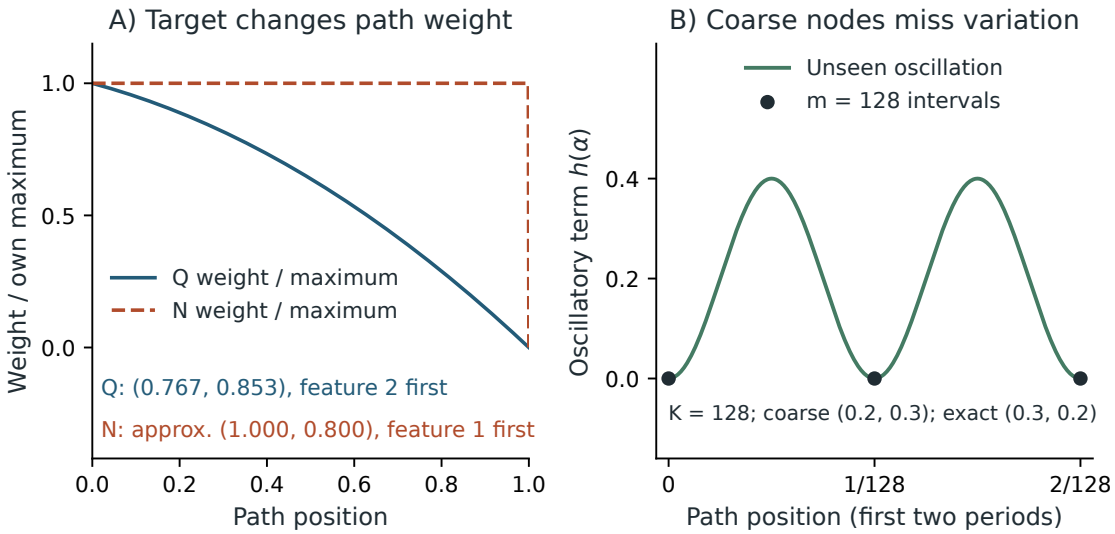


Figure 4 | Target weighting and unresolved variation can reverse feature order. A) The example uses $g(x_1, x_2) = x_1^2 + \frac{4}{5}x_2$, baseline $(0, 0)$, input $(1, 1)$ and reference $9/5$. Each target's weight on the policy gradient is scaled to its own path maximum. Blue is Q, orange N; annotated pairs are coordinate attributions. N's weight smoothly reaches zero in a narrow endpoint region (Appendix A.2). **B)** Part of $h(\alpha)$ for $K = 128$. Nodes for $m = 128$ intervals miss its oscillation; coordinate errors cancel in the completeness sum (Appendix B.1).

512 intervals. Its finest-budget ($m = 512$) completeness residuals are only 0.124% and 0.045%, so passing completeness at the finest budget does not establish coordinate convergence.

For state at the same episode's call 0, Q-target rankings at 256 versus 512 intervals have Spearman agreement 0.83; insertion AUC changes by 0.19 and deletion AUC by 0.10 under a common Q response. That insertion change is as large as the response-only shifts. Conversely, its call-12 Q-target state map changes by 32.1% between the two finest budgets with zero curve change. Even a 0.17% coordinate change between 256 and 512 intervals for PickCube episode 1, call 0, vision N moves Q-response insertion AUC by 0.0102, just above the 0.01 criterion. Integration error can alter evaluation outcomes or remain invisible to them. For a smooth action map, positive ϵ makes N's gradient continuous; these sampled budgets cannot establish an asymptotic order. Empirically, N residuals typically halve under doubling while Q residuals fall roughly fourfold, consistent with an unresolved narrow N transition.

The smooth aliasing construction in Appendix B.1 sets $h(\alpha) = \frac{2}{5}\sin^2(\pi K\alpha)$, $K = 128$, in a two-feature policy. Uniform grids with 32, 64 and 128 intervals give trapezoidal attributions $(1/5, 3/10)$ instead of exact $(3/10, 1/5)$ (Equations 22–23; Figure 4B). Equal and opposite errors cancel in completeness but cause 40% relative L_1 error and reverse the order. Appendix B.1's argument for m dividing K shows that the endpoint-inclusive rule also gives Equation 22. The inaccurate order scores better by insertion and deletion in this example (Appendix B.2). Thus even exact IG need not optimize perturbation scores, as the target-reversal example also shows. Budget and completeness checks detect problems but agreeing finite uniform grids cannot certify coordinates without assumptions controlling unresolved variation.

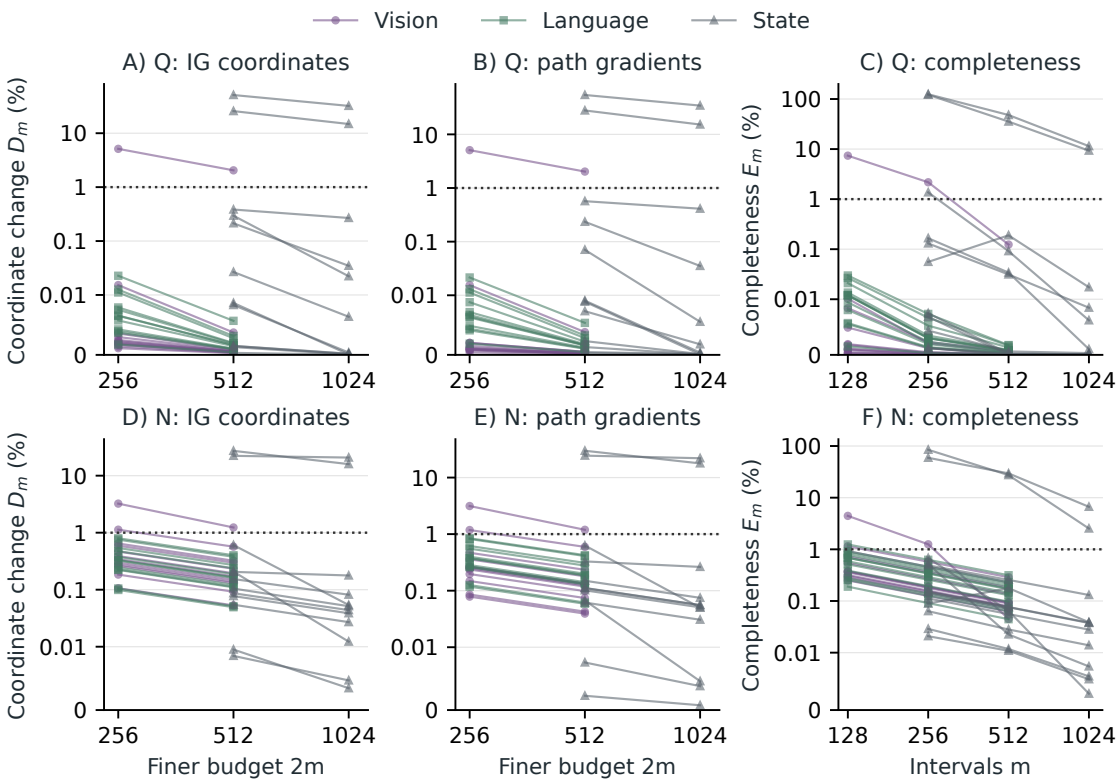


Figure 5 | Maps change with budget despite passing equality checks. Twelve completed RDT-170M contexts supply one line per context and modality. **A)–C)** Q target. **D)–F)** N target. Columns show IG coordinate change, mean path-gradient change, and completeness residual. Changes use the finer budget $2m$ in Equation 9; completeness uses m. Changes are shown only between adjacent budgets; three of the four vision N contexts with finest-pair violations fail only RMS-curve, RMS-area or AUC criteria, which are not plotted. The dotted line marks 1%. Purple circles denote vision, green squares language, and gray triangles state. The symmetric logarithmic scale is linear within $\pm 0.01\%$; the zero tick denotes an exact zero.

4 Discussion

For question one, response choice substantially changes fixed-ranking evaluations; overshoots can dominate means. For question two, finite numerical agreement does not certify integration. Question three, ranking utility, remains unresolved. Separately, attribution target changes can reverse exact feature order.

Normalization puts calls on a common scale but amplifies small all-baseline squared residuals. Hama and colleagues instead use unnormalized areas from the straight endpoint reference, avoiding this division while retaining response units[9]. On their full prefix grid, combining insertion with deletion oriented so larger means better gives zero random-order expectation even with interactions; either area alone need not. Their IG argument gives an optimal feature order for an increasing transform of a linear inner score, whose gradient direction is constant. Appendix A.2's nonlinear inner score falls outside that condition: feature 1 first is better under both responses, yet exact Q-target IG ranks feature 2 first.

4.1 A practical reporting procedure

1. Specify the attribution target separately from the scoring response, with the reference action, noise, feature groups, baseline and intervention schedule. The target-reversal example and response identity show why both choices matter.
2. Compare rankings in matched contexts under the same response, including response-matched random rankings or a simple fixed feature order. A common response change can reverse two rankings (Appendix A.4); justify the response or check both.
3. Keep interventions fixed when comparing responses. Report undefined normalization when $B = 0$, overshoots, distributions of B or unnormalized areas, medians, positive shares and full tails alongside means. For repeated calls from episodes, resample whole episodes.
4. Check repeatability, coordinate changes and completeness across budgets; separately report ranking and curve changes. Keep IG and mean path-gradient results distinct because multiplication by $x - x'$ can change rankings. Report the completed and planned context counts.

4.2 Limitations and next steps

The headline evidence is one simulated task, one model recorded as pretrained RDT-170M, two environment-seed blocks and 30 episodes per cohort, with no successes. A failing policy still tests a scoring-only identity, but its effect sizes and overshoot rates may not generalize. Existing fine-tuned RDT-1B PickCube runs succeed in 15 of 60 episodes. Rescoring their vision curves through the same checked procedure is the most direct available test with a more competent policy; additional tasks and physical robots would further test generality.

Normalized AUC is unbounded below at overshoots. No matched random or fixed-order control establishes ranking superiority, and no model-weight randomization test is performed[17]. The incomplete numerical sample has no completed RDT-1B context and cannot estimate how often numerical checks fail across policy contexts. It does not validate the original bf16 rankings: the saved PickCube runs lack original observations, noise arrays and exact checkpoint files. Future studies should preserve complete contexts and compare finer or adaptive integration against explicitly justified error controls.

4.3 Conclusion

N cannot increase either fully in-range AUC, and the saved PickCube evaluations show substantial response-only changes. Strict decreases favor deletion and disfavor insertion; absolute thresholds can give opposite verdicts. Overshoots can dominate means, especially through extreme first calls in vision insertion. Target choice can reverse exact IG rankings, and numerical checks reveal problems without certifying accuracy. Report targets and responses separately, compare rankings with matched controls, retain full distributions, and check numerical sensitivity.

Data and code availability

The accompanying archives contain saved curves, episode records, analysis and figure/table scripts, analytic checks, and exact file hashes. They reproduce results and manuscript without model execution. Original observations, noise and checkpoint files from the headline PickCube runs were not preserved. The separate numerical collection preserved noise; supplied records reproduce its diagnostic summaries. The public repository is https://github.com/itscool2b/ig-response-geometry; release manuscript-2026-10-04-polish contains this manuscript, its data, and the analysis and figure scripts.

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AI-use disclosure

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A Response identities and target-induced reversal

A.1 Normalized response difference

Assume exact self-reference and $B > 0$. Set $\eta = \sqrt{\epsilon}$, $b = \sqrt{B+\epsilon}$ and $z = \sqrt{r+\epsilon}$, so $B = (b-\eta)(b+\eta)$. Then

$$C_Q - C_N = \frac{b^2 - z^2}{B} - \frac{b-z}{b-\eta} = \frac{(z-\eta)(b-z)}{B}. \tag{12}$$

For $\eta \le z \le b$, the product is nonnegative and maximized at $z = (\eta + b)/2$, giving

$$0 \le C_Q - C_N \le \frac{(b-\eta)^2}{4B} = \frac{b-\eta}{4(b+\eta)} \le \frac{1}{4}. \tag{13}$$

For $z > b$, the difference is negative, so the $1/4$ upper bound holds for all $r \ge 0$. Nonnegative AUC weights sum to one, yielding $\Delta \le 1/4$ for every defined curve. On the nine-point schedule

they are $(0.005, 0.025, 0.045, 0.075, 0.10, 0.15, 0.225, 0.25, 0.125)$. The endpoints contribute zero difference and have total weight 0.13, giving $\Delta \leq 0.87/4 = 0.2175$. For positive ϵ and finite B, the fixed-B schedule bound $0.87(b-\eta)/[4(b+\eta)]$ is strictly smaller. Fully in-range curves additionally satisfy $\Delta \geq 0$, strictly if a point with positive AUC weight has $0 < r < B$. If $B = 0$, normalization is undefined. When the 1% and 5% language values coincide, dropping the 1% node changes their combined AUC weight by 0.02; the $1/4$ pointwise bound therefore bounds the paired change by $0.02/4 = 0.005$ for fully in-range calls.

A.2 Finite-stabilizer rank reversal

Let $g(x_1, x_2) = x_1^2 + \beta x_2$, with $\beta = 4/5$, $x' = (0, 0)$ and $x = (1, 1)$. Along the diagonal, the scalar action residual and its square are

$$\rho(\alpha) = (1-\alpha)(1+\alpha+\beta), \qquad r(\alpha,\alpha) = \rho(\alpha)^2.$$

Since $\nabla g = (2\alpha, \beta)$ there, differentiating the targets for $d_a = 1$ gives

$$\nabla Q = \rho(\alpha)(2\alpha, \beta), \qquad \nabla N = \frac{\rho(\alpha)}{\sqrt{\rho(\alpha)^2 + \epsilon}}(2\alpha, \beta). \tag{14}$$

Polynomial integration yields

$$A^Q = \left(\frac{1}{2} + \frac{\beta}{3}, \frac{2\beta}{3} + \frac{\beta^2}{2}\right) = \left(\frac{23}{30}, \frac{64}{75}\right), \tag{15}$$

so feature 2 leads by $13/150$. Norm attributions tend to $(1, \beta)$ as ϵ decreases to zero.

For the positive stabilizer, define $\psi(\rho) = 1 - \rho/\sqrt{\rho^2 + \epsilon} \geq 0$. Its derivative is $-\epsilon/(\rho^2+\epsilon)^{3/2} < 0$, and $\rho(\alpha) \geq \beta(1-\alpha)$. Therefore

$$\begin{aligned} \int_0^1 \psi(\rho(\alpha))\, d\alpha &\leq \int_0^1 \left[1 - \frac{\beta(1-\alpha)}{\sqrt{\beta^2(1-\alpha)^2 + \epsilon}}\right] d\alpha \\ &= 1 - \frac{\sqrt{\beta^2+\epsilon} - \sqrt{\epsilon}}{\beta} \leq \frac{\sqrt{\epsilon}}{\beta}. \end{aligned} \tag{16}$$

The identities $1 - A_1^N = \int_0^1 2\alpha\psi(\rho(\alpha))\, d\alpha$ and $\beta - A_2^N = \beta \int_0^1 \psi(\rho(\alpha))\, d\alpha$, together with $0 \leq 2\alpha \leq 2$ and $\epsilon = 10^{-12}$, give

$$0 \leq 1 - A_1^N \leq 2.5 \times 10^{-6}, \qquad 0 \leq \beta - A_2^N \leq 10^{-6}. \tag{17}$$

Thus $A_1^N - A_2^N \geq 0.1999975$. Every coordinate is positive, so ordinary and absolute-coordinate rankings agree. Only the target changes. At the half-feature insertion point, feature 1 first gives normalized Q response about 0.80 versus 0.69 for feature 2 first, and N response about 0.56 versus 0.44. Deletion swaps these intermediate values. Hence feature 1 first scores better under both responses, despite exact Q-target IG ranking feature 2 first.

A.3 Finite-grid random-order calibration

For $g(x) = \sum_{i=1}^n x_i$, an all-one input and zero baseline, every feature order gives residual distances k after k deletions and $n-k$ after k insertions. At $q = k/n$, normalized Q curves are $1-q^2$ and $1-(1-q)^2$. Both have second derivative -2, so their trapezoidal areas underestimate their common integral $2/3$ by $1/(6n^2)$ on the full-prefix grid. Their identical curvature also gives identical quadrature errors on any common grid, including an asymmetric one. The nine-point schedule with $n = 100$ gives area 0.65976. Unstabilized N gives the lines $1-q$ and q, both with trapezoidal area 0.5 on either schedule.

A.4 A common response change can reorder rankings

Take three features and insertion fractions $\{0, 1/3, 2/3, 1\}$, with $B = 1$. Let orders $(1, 2, 3)$ and $(3, 2, 1)$ give residual sequences $(1, 1/4, 1/4, 0)$ and $(1, 16/25, 0, 0)$, respectively. Their intermediate subsets are distinct, so these values can belong to one scalar policy with the same endpoints. Q insertion AUCs are $2/3$ and $31/50 = 0.620$; with $\epsilon = 0$, N AUCs are $1/2$ and $17/30 \approx 0.567$. Thus the better order reverses. The inequalities persist at $\epsilon = 10^{-12}$. This establishes possibility, not a measured comparison of the saved cohorts.

B A smooth counterexample to integration-grid agreement

B.1 Unresolved coordinate variation

For an integer $K > 0$, define

$$h(\alpha) = \frac{2}{5}\sin^2(\pi K\alpha), \tag{18}$$

$$g(x_1, x_2) = \frac{2}{5}x_1 + \frac{3}{5}x_2 + (x_1 - x_2)h\left(\frac{x_1 + x_2}{2}\right), \tag{19}$$

$$Q(x_1, x_2) = -\frac{1}{2}\big(g(x_1, x_2) - 1\big)^2. \tag{20}$$

Take baseline $(0, 0)$, actual input $(1, 1)$, and fixed reference $g(1, 1) = 1$. The transverse term vanishes along (α, α), so $g(\alpha, \alpha) = \alpha$ and $Q(\alpha, \alpha) = -(1 - \alpha)^2/2$. Differentiating before restricting to that path gives

$$\nabla Q(\alpha, \alpha) = (1 - \alpha)\left(\frac{2}{5} + h(\alpha), \frac{3}{5} - h(\alpha)\right). \tag{21}$$

Thus the oscillation transfers gradient between coordinates while leaving their sum unchanged.

Use the composite trapezoidal rule with m intervals. Whenever m divides K, every node $\alpha = j/m$ has $h(\alpha) = 0$. The sampled coordinate functions are then linear. Both the trapezoidal rule and endpoint-inclusive average integrate a linear function exactly on any uniform grid, giving

$$A_m^{\text{trap}} = \left(\frac{1}{5}, \frac{3}{10}\right), \qquad \mathbf{1}^\mathsf{T} A_m^{\text{trap}} = \frac{1}{2} = Q(1, 1) - Q(0, 0). \tag{22}$$

Here $\mathbf{1} = (1, 1)^\mathsf{T}$ sums the coordinates. Every such grid has zero completeness residual and ranks feature 2 first, giving identical masks and perturbation curves under a deterministic ranking rule.

The exact coordinate integrals differ. Since $h(1 - \alpha) = h(\alpha)$, substitution gives

$$\int_0^1 \alpha h(\alpha)\, d\alpha = \int_0^1 (1 - \alpha)h(\alpha)\, d\alpha.$$

Adding these expressions and using $\int_0^1 h(\alpha)\, d\alpha = 1/5$ shows that either integral is $1/10$. Consequently,

$$A_{\text{exact}}^Q = \left(\frac{3}{10}, \frac{1}{5}\right). \tag{23}$$

The exact order puts feature 1 first. Coarse errors $(-1/10, 1/10)$ cancel in completeness, but their L_1 norm, $1/5$, is 40% of the exact attribution's norm.

On the $2K$ grid, h alternates between zero and $2/5$, with trapezoidal integral $1/5$. Reflection symmetry gives area $1/10$ for $(1 - \alpha)h$, recovering Equation 23. This recovery is construction-

specific. For finitely many uniform budgets, a common multiple K hides every oscillation. Agreement cannot certify coordinate integrals without a regularity bound on unresolved variation.

The endpoint-inclusive average has the same coarse failure. When m divides K, the average of the sampled linear functions also yields Equation 22. On the $2K$ grid, only odd nodes contribute to $(1-\alpha)h$. Their sum is $\frac{2}{5}\sum_{\ell=0}^{K-1}(1-(2\ell+1)/(2K)) = K/5$. Dividing by the $2K+1$ nodes gives

$$A_{2K}^{\text{avg}} = \left(\frac{1}{5} + \frac{K}{5(2K+1)},\ \frac{3}{10} - \frac{K}{5(2K+1)}\right). \tag{24}$$

The first coordinate exceeds the second for $K > 1/2$, including every allowed integer. Coordinates remain inexact; exact $2K$ recovery belongs to the trapezoidal rule.

B.2 Exact IG and perturbation quality

Set $K = 128$, so $h(1/2) = 0$ and the whole-feature interventions give $g(1,0) = 2/5$ and $g(0,1) = 3/5$. Under the same normalized Q response and fractions $\{0, 1/2, 1\}$, feature-2-first insertion values are $(0, 0.84, 1)$ and deletion values are $(1, 0.64, 0)$. Their trapezoidal AUCs are 0.67 and 0.57. The exact-IG feature-1-first order instead gives insertion AUC 0.57 and deletion AUC 0.67. The inaccurate order therefore scores better by both conventional criteria. The transverse term alters coordinate gradients along the diagonal without altering these finite off-path intervention values. Exact path integration and perturbation-score optimization are distinct questions.

CPU fp32/fp64 checks at $K = 128$ verify trapezoidal coordinates, completeness, repeatability and responses: budgets 32, 64 and 128 alias; 256 recovers the analytic order to numerical tolerance. The endpoint-inclusive result is analytic only; these checks use no RDT data.